

① Informed v/s Uninformed

② Game Optimal

→ Heuristic & Factors

Heuristic can be defined as the estimation or approximation of distance from the current node to the goal node. (closeness to the goal)

Conditions for best heuristic $h(n)$:

1. Admissibility

$$h(n) \leq h^*(n)$$

$\uparrow$ heuristic cost $\uparrow$ estimated cost

The heuristic funcn. never overestimate the cost to reach the goal.

Eg: straight line distance

2. Consistency

$$h(n) \leq c(n, a, n') + h(n')$$

$\nwarrow$ n to goal $\nwarrow$ step cost b/w n & n' $\nwarrow$ estimated cost from n'

For every node n & successor n' of n generated by a , the estimated cost of

reaching goal from is the step cost from $n \rightarrow n'$ & the estimated cost from n' to goal.

3. Computational Simplicity - heuristic should be easy & fast to compute.

→ Describe Hill Climbing with problems + solutions

Hill Climbing problem is a local search algorithm which is simply loop that moves continually upwards, in the direction of increasing value.

→ It terminates at the peak where no neighbour has a higher value.

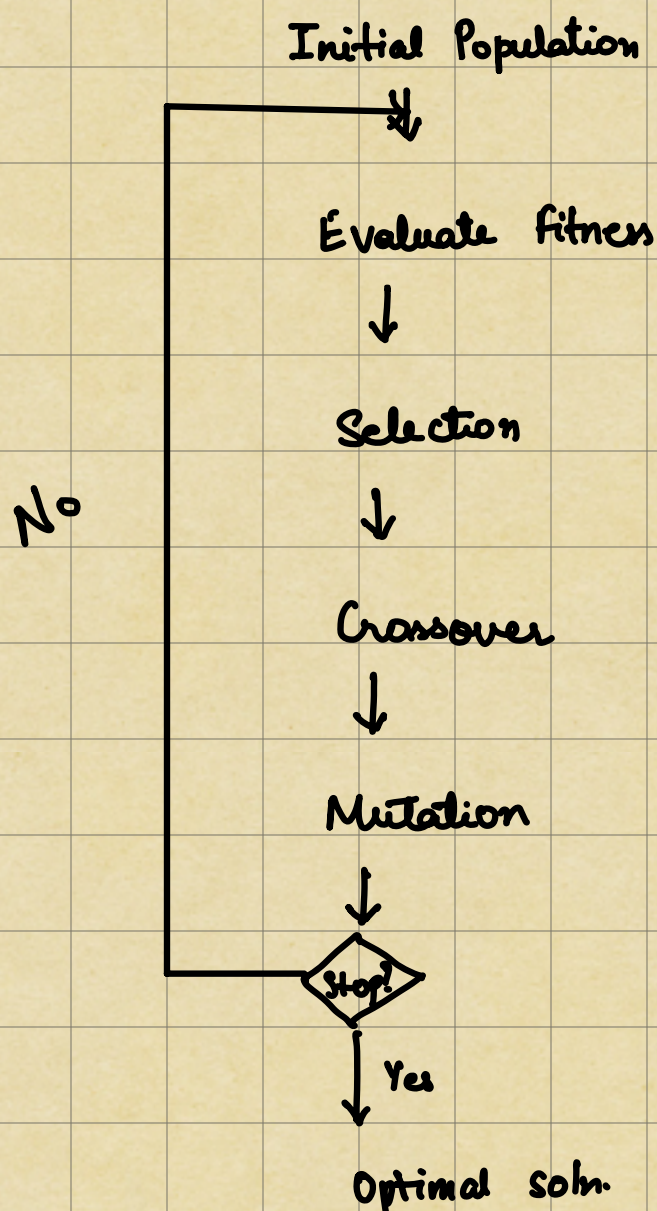
→ Does not maintain a search tree, so the ds for the current node only needs the state & value of obj. function.

→ It does not look beyond the immediate neighbours of the current state.

Problems	Solution
Local maxima: a state better than its imm. neighb. but a better state exists	Backtrack to some earlier node & try a new direction
Plateau: a flat region in state space where neighb. have same value $\Rightarrow$ search might get lost	Make a big jump or apply the operators again & again though situation continues
Ridge: a region higher than neighbours but has a slope itself (special local maxima)	Apply two or more operators before doing the test; will help in moving several dirn. at once.

→ Genetic Algo + terms + Flowchart

Genetic Algorithms are an abstraction on the real biological evolution. It is used for solving complex problems such as NP Hard problems with focus on optimization.



Chromosome: A set / seq of parameters that represents
a ^{potential} soln. for a problem at hand.

↓
Each individual posn = Gene

Fitness Function: competing ability or quality of chromosome
in the population $\Rightarrow$ $\uparrow$ fitness value $\uparrow$ Selection

Selection: fittest individuals pass to the next generation

Crossover: is a genetic operator used to combine genetic
info of two parents to generate new offspring.

child = best of two traits

↙
Single point

↘
Two-point

Mutation: is a genetic operator used to maintain genetic diversity
within a population from one generation to another

↓
w/o $\Rightarrow$ similar $\Rightarrow$ local optimum

Alpha - Beta Pruning & Min-Max

- Min-Max algo is an adversarial search algo that aims to find mx value to win game
- Uses DFS, and propagates Min-Max values up the tree.

Complete - If tree is FINITE

$$TC - O(b^m)$$

$$SC - O(bm)$$

→ Alpha Beta pruning improves on min-max

↳ involves two threshold values

α → max value along maximizer $(-\infty)$

β → min value " minimizer $(+\infty)$

α = MAX player

β = MIN player

Only sent to the child nodes.